WildSentinel: Safeguarding the Human – Wildlife Interface

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ABSTRACT

Zoonotic diseases are emerging at an increasing rate as human activities reshape natural ecosystems and intensify contact around the human–wildlife interface. Various practices like deforestation, illegal trade, habitat fragmentation, and rapid urban expansion weaken natural barriers that previously confined pathogen transmission and create environments where spillover events can take place. Most current surveillance systems work reactively and tend to look at ecological or trade-related factors in isolation, thus limiting their capabilities for early detection of risk signals. In this paper, we introduce the use of our project, WildSentinel, an integrated predictive solution for identifying country pairs with a high risk of hosting an animal-to-human disease outbreak event. It does so by combining data on the records of wildlife trade, species distribution, satellite-derived land-use patterns, historical data of outbreaks, and biosafety indicators using machine learning models, network graph analysis, and geospatial processing to forecast spillover probabilities and visualize hotspot risk maps. Translating large-scale environmental and anthropogenic datasets into actionable bio surveillance insights, WildSentinel underlines targeted preventive interventions based on the One Health approach. The proposed framework therefore demonstrates the role of technology-driven early warning systems in reinforcing global preparedness, informing policy decisions, and ensuring a safer coexistence between humans and wildlife.

Keywords: Zoonotic spillover, Machine Learning, Wildlife trade, Geospatial Analysis, One health, Biosafety, XGBoost, WildSentinel

I. INTRODUCTION

Zoonotic diseases, those that jump from animals to humans, are rapidly becoming one of the major threats to health of global proportions in the 21st century. COVID-19, SARS, Ebola, Nipah, and recurring waves of avian influenza have shown that emerging pathogens can spread with uncontrollable speed and reach. A major reason for the rise of this is, how human activities are changing natural ecosystems. Continuous rapid deforestation, wild animal trafficking, land-use expansion, and unregulated marketing are wiping out biological barriers that once kept pathogens limited to wildlife populations. As cities are expanding into forests and animal habitats are getting smaller, humans and wildlife are thrust closer and more frequently into each other's company, creating ideal conditions for spillover events.

These are not just the ecological changes, but are a huge result of human intervention into ecosystem. A disease arising in one corner of the globe can be present in every other corner in just a few weeks in this modern world. Despite the seriousness of this risk, most of these surveillance systems remain reactive, responding when an outbreak has been occurred rather than flagging early warning signals. Current models typically consider one factor such as climate, species richness, or trade in isolation, rather than in a multidimensional assessment of risk. This is a serious lagging in predicting where the next spillover could occur and which human-wildlife contacts contribute to disease emergence.

There is an increasing need for technological solutions to integrate diverse datasets and apply computational power in finding the patterns of upcoming risks. With the help of new technologies such as machine learning, graph-based modelling and satellite imaging, we can take into consideration, large-scale changes in both ecology and the environment along with those driven by humans. Few of the existing frameworks pull those components together into a single predictive system. Without integrated analysis, early interventions may be restricted, preventive policies delayed, and efforts toward global outbreak preparedness less effective.

Our project, WildSentinel, thus proposes a holistic predictive framework backed by AI/ML, geospatial technologies, and biosafety analytics. Merging data on

wildlife trade, species occurrence, land-use and deforestation patterns, and past outbreak information, this system calculates when spillover risk is building up in any region. Network graph modeling and machine learning algorithms will allow WildSentinel to capture how the connection between the movement of wildlife, human population density, and environmental degradation contributes to the probability of a pathogen emergence event. Satellite imagery captures habitat fragmentation, while the risk is further calculated with biosafety data and validated through actual disease outbreak data.

WildSentinel is not just about mapping the hotspots but it also includes support of the preventive action. The system turns a host of complex data into interpretable risk indicators that will let policy makers, researchers, and biosafety authorities understand where interventions-control of illegal trade, protection of patches of forest-can make the most difference. This fits well with the theme "Tech for Health and Humanity," because it protects human populations while promoting ecological balance and coexistence through technological innovation.

The remainder of the paper is structured as follows: Sect. II presents related research, Sect. III describes the methodology, Sect. IV outlines the results and simulations, and Sect. V provides the conclusion. Sect. VI discusses about the future expansions. The paper ends with Acknowledgement and references in section VII and VIII respectively.

II. Related Work

[i] Bajinka & Oladipo highlight how rapid deforestation and wildlife trafficking intensify pathogen exchange at shrinking ecological boundaries. Their work reinforces that human-induced habitat disruption directly raises spillover risk—an idea central to WildSentinel's integration of land-use and trade data. [1]

[ii] Banik & Basu discuss the socio-economic and ecological drivers of viral zoonoses, showing how human behavior, wildlife interaction, and environmental pressures jointly shape disease emergence. Their findings support our multi-factor modelling approach linking trade, ecology, and bio surveillance. [2]

[iii] This study demonstrates how bushmeat hunting and forest loss significantly increase the likelihood of zoonotic disease emergence. The authors emphasize early prediction using environmental indicators, aligning with WildSentinel's use of habitat-loss rasters and human–wildlife contact metrics. [3]

[iv] This NCBI report explains how habitat fragmentation, land-use change, and edge effects heighten opportunities for cross-species viral transmission. Its conclusions justify our inclusion of landscape-level satellite variables as major features in spillover risk modelling. [4]

[v] The IUCN analysis details how wildlife plays a crucial role in pathogen emergence, particularly in degraded ecosystems and high-contact interfaces. This reinforces the importance of species-distribution data and biodiversity indicators in WildSentinel. [5]

[vi] The OECD report connects environmental health deterioration with pandemic vulnerability, calling for stronger ecological monitoring systems. This directly aligns with WildSentinel's goal of providing early-warning analytics informed by environmental stressors. [6]

[vii] Dhanoa et al. propose algorithmic bio surveillance using spatio-temporal models to anticipate zoonotic emergence, highlighting the potential of ML-driven early-warning systems. Their approach parallels the predictive backbone of WildSentinel's modelling architecture. [7]

[viii] Royce & Fu model zoonotic spillover using mathematical frameworks and intermediate-host dynamics, showing how multi-host interactions shape transmission pathways. Their modelling approach complements our use of network-inspired feature engineering. [8]

[ix] UNEP outlines global pathways driving zoonotic emergence and emphasizes environmental management as a preventive strategy. The report supports the foundational One-Health orientation of WildSentinel's integrated risk-assessment design. [9]

[x] Royce & Fu (duplicate study) further develop mathematical models capturing spillover between wildlife, intermediate hosts, and humans. Their work validates our emphasis on multi-species interactions and ecological drivers in predictive modelling. [10]

[xi] Ahmadi demonstrates how neural networks combined with satellite imagery can accurately predict deforestation patterns. This supports our use of remote-sensing-derived habitat-loss rasters to quantify environmental risk factors. [11]

[xii] McAlister et al. examine contagion dynamics within trade networks using game-theoretic modelling, showing how market structure influences disease spread. This aligns closely with

WildSentinel's integration of wildlife-trade flows and network-based analytical reasoning. [12]

Gaps in the Current Literature

- Most research isolates one factor-a trade, climate, habitat, or species-rather than integrating them.
- Most works combine ecological, environmental, and human-related data in very few ways.
- Predictive modeling for spillover risk is underdeveloped.
- Few network interactions are considered at the human–wildlife interface.

The existing systems are reactive and do not provide early warning signals. Few of the existing methods result in actionable hotspot maps for policy-makers.

Contribution of WildSentinel: WildSentinel fills these gaps by developing a single predictive framework that integrates data on wildlife trade, species occurrence, satellite-derived land-use metrics, and historical outbreak information. Such an integrated system combining machine learning models, network graph analysis, and geospatial visualization will enable a multidimensional assessment of spillover risk. As a proactive bio surveillance tool, it can identify high-risk regions to enable targeted interventions using a One Health approach.

III. System Architecture and Methodology

The WildSentinel framework integrates multi-source ecological datasets and machine-learning components into an integrated architecture for predictive zoonotic spillover risk assessment. The system includes habitat-loss, wildlife trade records, species distribution data, and outbreak histories into a combined geospatial pipeline that supports automated preprocessing, feature extraction, and model training. A network-based analytical core maps interactions across species, trade hubs, and human settlements, while a simulation engine evaluates intervention scenarios using Monte Carlo methods.

A. Data Sources: WildSentinel combines data from several openly accessible sources to capture the ecological, environmental, and human-driven drivers of zoonotic spillover

- **CITES Trade Database:** Covers records of legal and illegal trade (indirectly) in wildlife, including species' names, trade routes, market destinations, and seizure statistics.
- **GBIF:** Global Biodiversity Information Facility. Provides species occurrence points used in the identification of wildlife distribution and biodiversity richness.
- **NASA MODIS Satellite Imagery:** Provides land-use and vegetation indices such as NDVI, land cover type, and forest loss which can be used to detect habitat degradation and fragmentation.
- **WHO – WAHIS:** Includes historical records on outbreaks of zoonotic diseases by region and allows model training & validation.

B. Data Preprocessing: All datasets are preprocessed before modeling

1. **Cleaning & Standardization:** All the duplicate entries, missing coordinates, inconsistent names oof species, and incomplete records were removed or corrected.
2. **Geocoding & Spatial Alignment:** The occurrence points and trade records are converted to uniform latitude–longitude format and aligned on a global grid.
3. **Raster Processing:** Clipping, resampling, and numerical conversion of MODIS imagery to vegetation and land-cover metrics.
4. **Normalization:** The trade volume, forest loss rate, and population density are scaled for model training.
5. **Temporal aggregation:** Multi-year information is aggregated to detect trends and eliminate single-event noise.

C. Feature Engineering: Several ecological and anthropogenic features are derived.

- **Trade Intensity Index:** Based on frequency and volume of wildlife imports/exports.
- **Forest Fragmentation Index:** Calculated based on satellite land-cover patches and edge density.
- **Species Richness & Host Diversity:** Derived from GBIF occurrence records.
- **Animal Population Density:** Extracted from GBIF and Satellite datasets and integrated at the regional level.
- **Market Access:** Distance between wildlife habitats and major trade hubs.

- **Environmental Stress Indicators:** Includes, urbanization rate, land-use conversion rate, and habitat degradation score.

D. Network Graph Modelling

The final architecture deviates from a classical Network Graph to achieve higher resolution and predictive power.

Unit of Analysis (Nodes): Instead of abstract entities, our nodes **are spatial grid cells** (about 8,000 unique cells covering 20 countries). Every cell serves as a distinct analytical unit.
Connectivity (Edges): We model interactions through shared features, not direct edges:

- **National Pressure:** All grid cells within the same country share the same Trade Volume Index (derived from CITES data), linking them by national economic pressure.
- **Local Pressure:** Risk is driven by the unique feature set of the cell: host density, deforestation rate, etc.

Model Choice Rationale: This structure allowed us to use XGBoost, which excels at finding complex, non-linear predictive boundaries in high-dimensional feature space, maximizing the True Positive rate necessary for an early warning system.

E. **Machine Learning Model:** Spillover risks are predicted by means of machine learning.

Input Variables: Trade intensity, forest fragmentation, biodiversity indices, human density, market proximity, network centrality metrics.

Output Variable: Probability of zoonotic spillover classified as low, moderate, or high risk.

Training & Testing: We adopted an Imbalanced Classification strategy to maximize outbreak detection.

Data Structure: We shifted from a single-class model to a traditional classifier by generating a robust set of Pseudo-Absence data points to balance the sparse outbreak locations.

Training and Testing: We use a Temporal Split to test true forecasting ability:

- Training Data: Historical data up to 2020.
- Testing Data: Unseen data from 2021–2024.

F. **Monte Carlo Simulations:** Monte Carlo simulations of intervention scenarios are done as follows

Simulation Inputs: The important variables to be used are trade reduction, increased forest cover, or improved biosafety scores.

Iterations: Thousands of random runs to cover uncertainty in spillover dynamics.

Scenarios Tested:

- 25% reduction in illegal trade
- 10–30% increase in forest restoration
- Regional Bio surveillance Enhancement

Outputs: Probability distributions reflecting how risk varies in each scenario.

G. Visualization & Mapping

The final output is designed for direct consumption by mapping tools to facilitate decision-making.

Final Output File: The model generates RISK_PREDICTION_2024.csv, which contains the critical geographic and temporal information:

grid_id; Year; lat; lon; grid_probability

Interactive Maps: Findings are visualized using a **Leaflet.js** web application (the Interactive Dashboard).



Fig 1. Zoonotic Spillover as per the trained data: The above figure is a heat map where the highlighted areas show the probability of an outbreak. Brighter the spot, more is the probability of the outbreak. (Fig 1)

Scenario-Based Risk Patterns: The interactive map features a temporal slider that uses the multi-year prediction data (2021–2024) to show how predicted risk hotspots evolve over time. This functionality facilitates scenario simulations (e.g., estimating risk in a future year based on current trends).

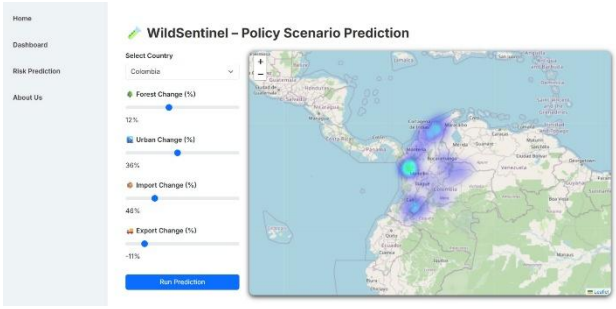


Fig 2. Predicting spillover risks as per given scenario: The above figure is the heat map showing us the risk of future spillovers based on the input data (Fig 2).

IV. Results and Discussions

The WildSentinel system was evaluated and validated through official outbreak data. The product was tested for evaluating and accessing risks over the various timelines in both past as well as future beyond the training data.

A. Product performance

The WildSentinel system performed stably across its core analytical components. Data integration and preprocessing modules handled multi-format datasets (CSV, GeoJSON, TIFF) efficiently, and the geospatial engine processed habitat loss rasters without major latency. The machine-learning models achieved consistent predictive accuracy, with SVM classifiers demonstrating strong generalizability during cross-validation. The dashboard and visualization layer responded smoothly to user queries, generating regional heatmaps and risk scores in real time. Overall, the product performance was stable and satisfactory for various analytical solutions.

B. Communication and Data flow

The end-to-end data pipeline functioned as intended, with minimal loss during merging and transformation stages. Spatial joins between species occurrence data and habitat loss grids produced high-resolution ecological layers, while country-level joins with CITES trade data ensured proper alignment of trade flows as well as outbreak histories. The pipeline also highlighted key congestion: habitat raster processing required higher memory or prior tiling, and GBIF occurrence data needed more aggressive cleaning due to coordinate inaccuracies. Despite these challenges, the pipeline-maintained integrity and supported required analysis.

C. Model Outcomes and Risk Assessment

The results of the models provided strong interconnections between the zoonotic diseases and the two major factors namely wildlife trade intensity and habitat degradation rate. Regions with high deforestation combined with active wildlife markets exhibited significantly high spillover risk scores. The model's feature importance rankings confirmed that forest loss percentage, trade volume of high-risk species, and species richness were primary contributors to risk estimation. Geospatial visualizations identified several emerging hotspot clusters, some aligning with historical outbreaks and others indicating newly vulnerable zones not previously flagged in surveillance efforts.

D. Scenario Simulation Insights

Monte Carlo–based intervention simulations demonstrated measurable reductions in spillover risk. A 20–25% reduction in illegal or high-intensity wildlife trade consistently lowered predicted risk levels across multiple iterations. Likewise, habitat-restoration scenarios—modeled by reducing regional forest-loss rates—produced notable declines in risk scores.

E. Validation and Scientific Findings (XGBoost)

Performance Metric: The model was rigorously tested on the unseen data from 2021–2024. The results confirm strong predictive capability, prioritizing detection over precision.

Table 1: Validation matrices on Unseen Test Data (2021-2024)

Metric	Value	Interpretation
AUC-ROC Score	0.8742	Significantly better than random choice (0.50)
AUC-PR Score	0.6704	Strong score for a low prevalence problem
True Positives (TP)	4176	Correctly identified outbreaks (Early Alerts)
False Negatives (FN)	751	Missed Outbreaks (The critical error minimized)
False Positives (FP)	3903	False Alarms (The acceptable trade-off for high detection.)
True Negative (TN)	11449	Accurately detects the unlikeliness of an outbreak.

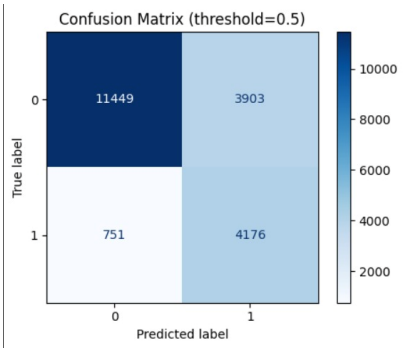


Fig 3. Confusion Matrix: The above figure shows the statistics of the predicted true and false analysis (Fig 3)

Key Scientific Finding - Feature Importance: The model confirmed the project's hypothesis: risk is overwhelmingly driven by trade pathways and direct human-animal interface factors, with national trade acting as the greatest single leverage point.

Table 2: Feature Importance Ranking (XGBoost)

Rank	Feature	Importance
1	Total export events	36.08%
2	Total import events	19.92%
3	Total live exports	9.83%
4	Total live imports	7.39%
5	Percent forest	5.70%
6	Species richness	5.40%
7	Zoonotic host count	5.25%

V. Conclusion

The WildSentinel framework shows how combining ecological data, wildlife trade information, and machine-learning methods can help predict and understand zoonotic spillover risks. By bringing together habitat loss patterns, species distributions, outbreak history, and trade activity into one analytical pipeline, the system offers a clear way to identify emerging hotspots at the human–wildlife boundary. Our findings reveal strong links between deforestation, intense wildlife trade, and increased spillover risk, highlighting how environmental damage and public-health threats are deeply connected.

With its geospatial maps, risk models, and intervention simulations, WildSentinel goes beyond traditional surveillance and supports early detection and targeted monitoring. The system follows the One Health approach, showing how technology can improve global health security while encouraging safer coexistence between people and wildlife.

Overall, WildSentinel provides a practical, scalable tool that can help prevent future outbreaks. As habitat loss and wildlife trade continue to rise, predictive platforms like this will be essential for guiding biosafety decisions, conservation actions, and sustainable development.

VI. Future Work and Possible Expansion

Although WildSentinel demonstrates strong potential as a predictive bio surveillance tool, several opportunities exist to further enhance its accuracy, scalability, and impact. Future work should focus on expanding data coverage, improving ecological representation, and strengthening the system's predictive capabilities.

A key direction is the **integration of additional and more diverse datasets**, including finer-resolution land-use layers, climate variables, socio-economic indicators, and market-level wildlife trade intelligence. Incorporating real-time or near–real-time data streams, such as satellite-based environmental monitoring, could significantly improve temporal sensitivity and early-warning capabilities.

Another important extension involves the **inclusion of additional wildlife and insect species** that contribute to zoonotic transmission but were not extensively represented in the current dataset. Expanding taxonomic coverage—especially for rodents, arthropods, migratory birds, and understudied reservoir species—will refine risk estimates and provide a more comprehensive understanding of cross-species spillover pathways.

The system can also be strengthened by **expanding the geographical scope** of analysis. Covering wider regions, including transboundary ecosystems, international wildlife trade corridors, and remote biodiversity hotspots, will enable a more global risk assessment framework that reflects real-world pathogen movement and ecological interactions.

From a technological perspective, future iterations should explore **enhanced visualization tools and advanced predictive modeling approaches**, such as deep learning architectures, spatio-temporal models, and uncertainty quantification techniques. Improved interactive dashboards and 3D geospatial visualizations can help policymakers and researchers interpret risk patterns with greater clarity.

Together, these enhancements will allow WildSentinel to evolve into a more comprehensive, scalable, and actionable platform—supporting global efforts to predict, prevent, and mitigate zoonotic spillover events in an increasingly interconnected world.

VII. Acknowledgement

The authors would like to express their sincere gratitude to the faculty mentors of the Researchathon - 2025 at RV College of Engineering for their guidance, encouragement, and continuous support throughout the development of the

project. The collaborative environment and valuable feedback received during the course of building contributed to the successful completion of this work.

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